**4** **(a)** Fig. 4.1 shows a piece of metal, of mass 50 g, held in the flame of a Bunsen burner for several minutes. The metal is then quickly transferred and immersed in 130 g of water contained in a calorimeter.

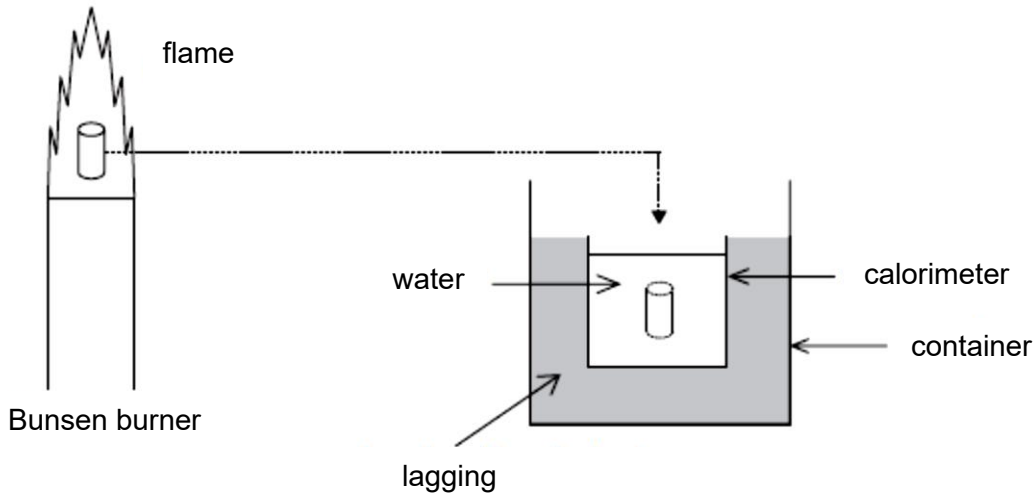


**Fig. 4.1**

The water into which the metal has been placed is stirred until it reaches a steady temperature. The following data are available:

| | |
|---|---|
| heat capacity of metal | 82.7 J $K^{-1}$ |
| specific heat capacity of the water | $4.2 \times 10^3$ J $kg^{-1}$ $K^{-1}$ |
| heat capacity of the calorimeter | 54.6 J $K^{-1}$ |
| initial temperature of the water | 25 °C |
| final temperature of the water | 90 °C |

Use the data to calculate the temperature of the Bunsen flame and state an assumption made for your calculation.

temperature = ..... °C

Assumption: .....

..... [3]

**(b)** The gas in the cylinder of a diesel engine can be considered to undergo a cycle of changes of pressure, volume and temperature. One such cycle, for an ideal gas, is shown in Fig. 4.2. Processes A to B and C to D take place without heat exchange with the surroundings.

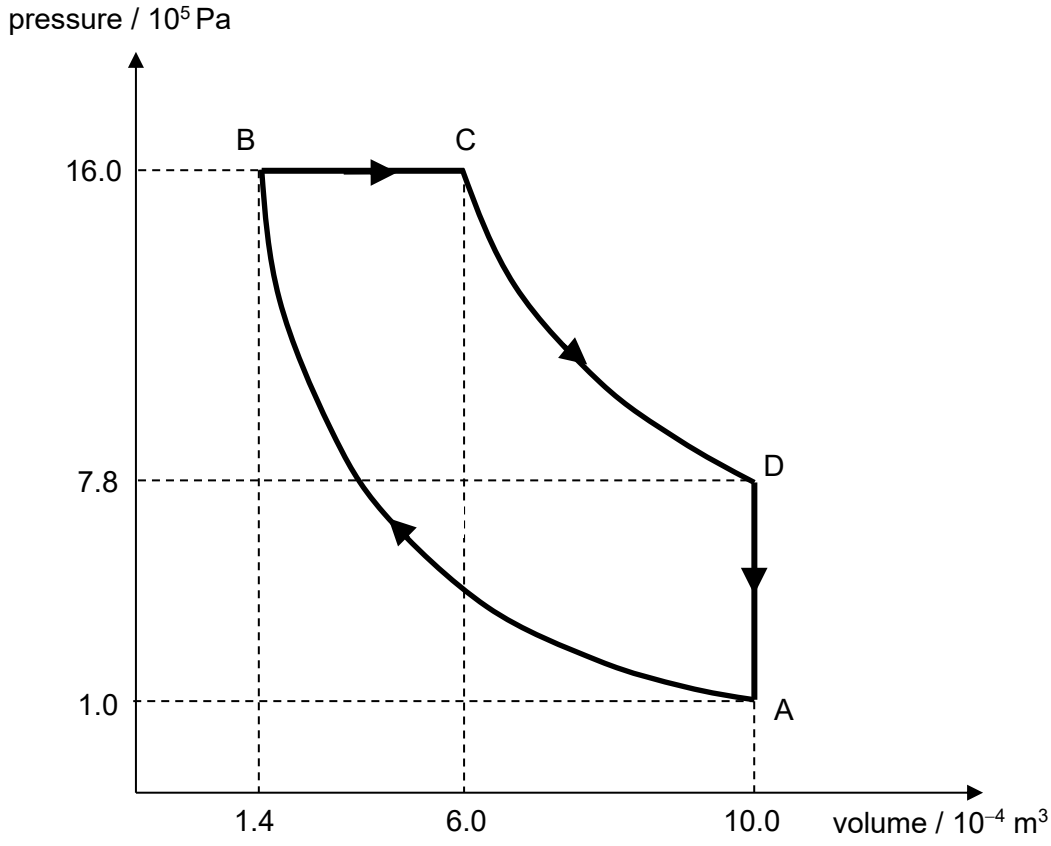


**Fig. 4.2**

Complete the table below.

| Process | Heat supplied to gas / J | Work done on gas / J | Increase in internal energy of gas / J |
|---|---|---|---|
| A to B | | 300 | |
| B to C | 2580 | | |
| C to D | | –440 | |
| D to A | –1700 | | |

[4]

**(c)** A fixed mass of ideal gas is heated from temperature $T_1$ to $T_2$ at constant volume. Explain why a greater amount of heat is required to heat the same mass of ideal gas from $T_1$ to $T_2$ at constant pressure.

.....

.....

.....

..... [3]